



CESAB
CENTRE DE SYNTHÈSE ET D'ANALYSE
SUR LA BIODIVERSITÉ

Introduction to AI techniques in evidence syntheses

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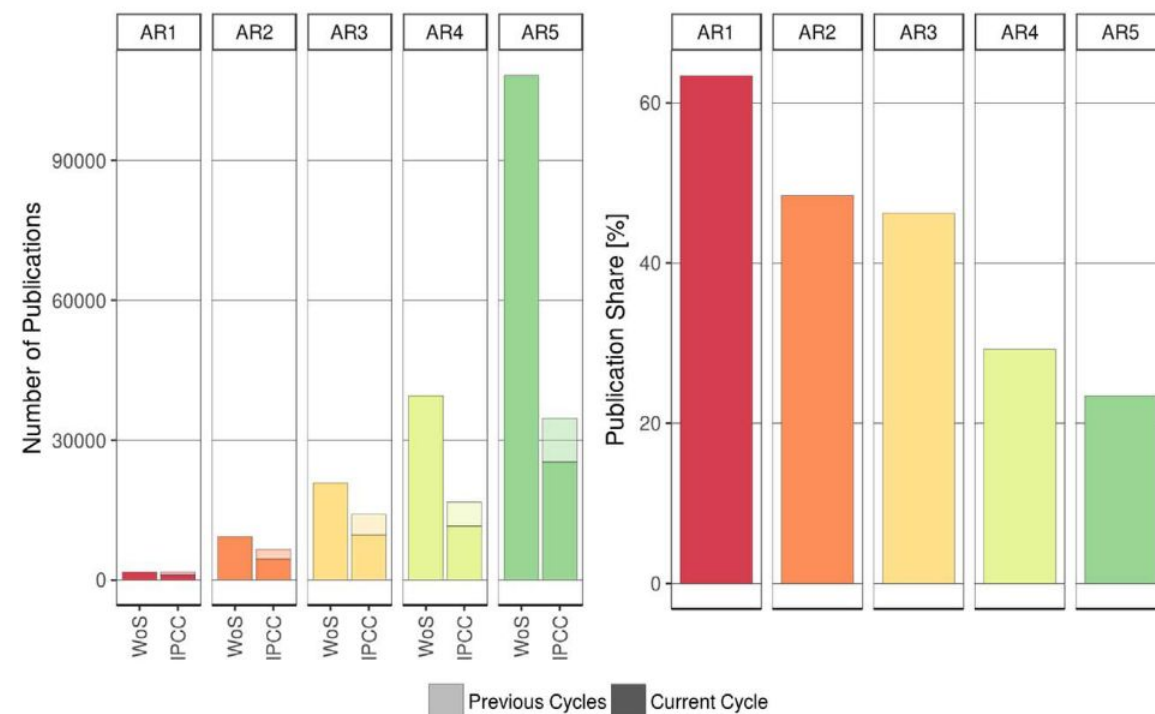
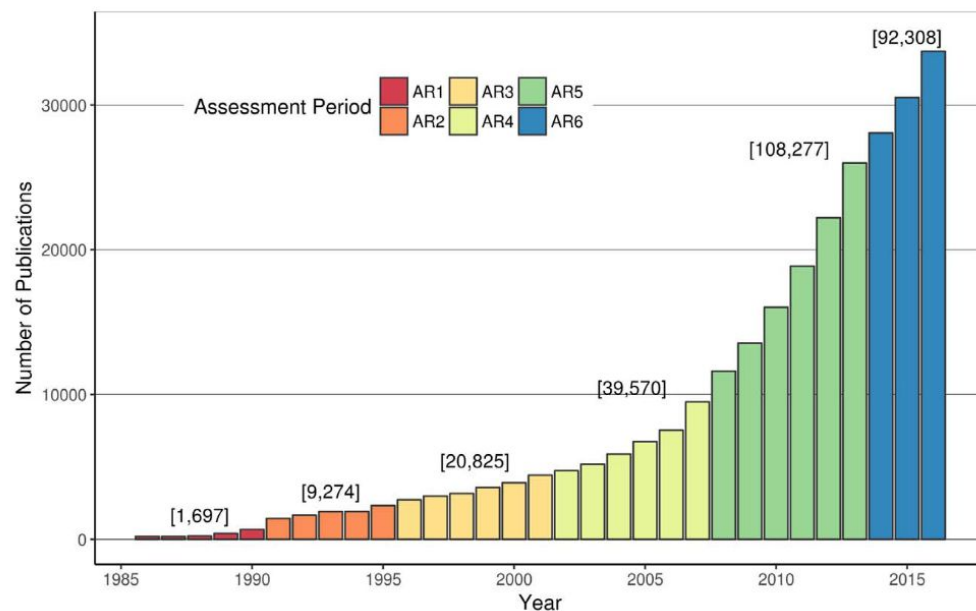
Outline

1. What is the need?
2. Portfolio of tools, applications & limitations
3. Which tool to use? And how to evaluate and report its use? The RAISE guidelines

What is the need ?

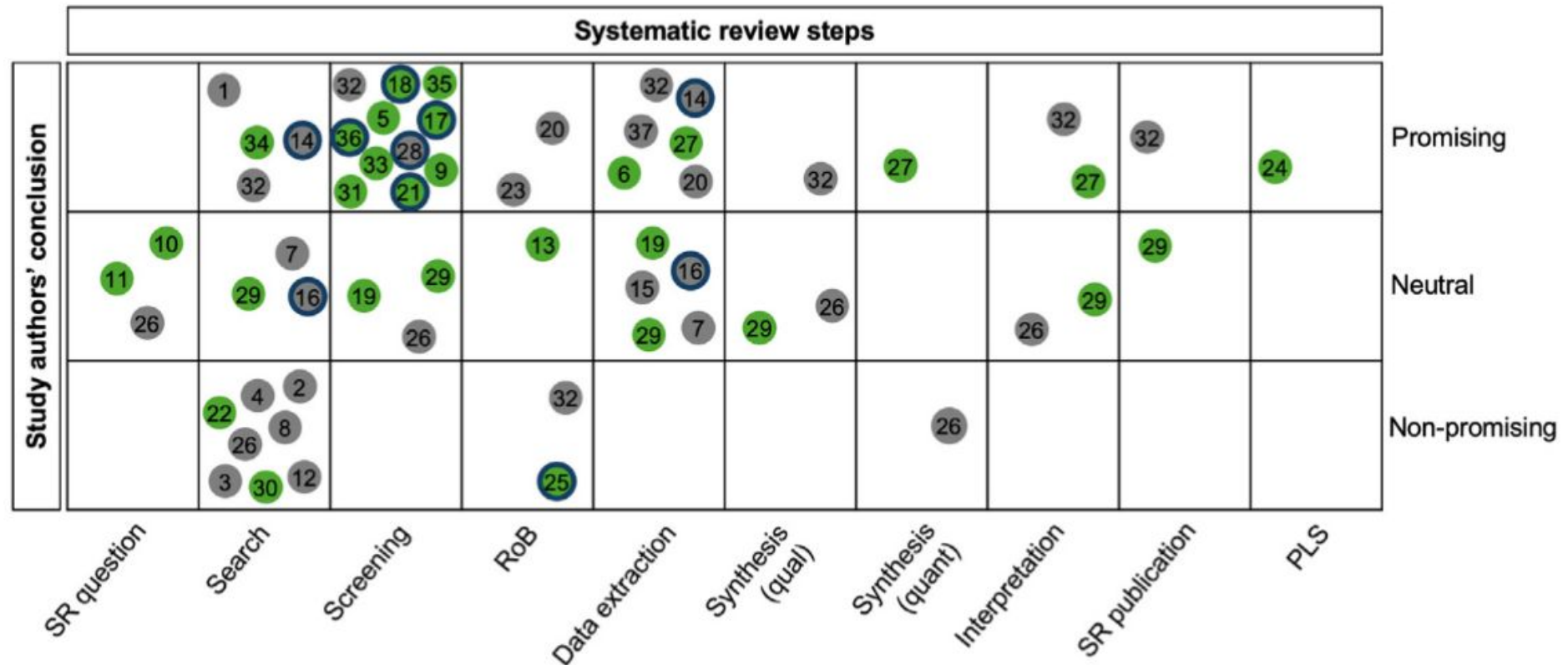
Publications is exponentially increasing

What are the implications for evidence syntheses?



Source: Minx et al. 2017

Emergence of LLM applications in evidence syntheses, some show promise



Source: Lieberum et al., *Journal of Clinical Epidemiology*, 2025



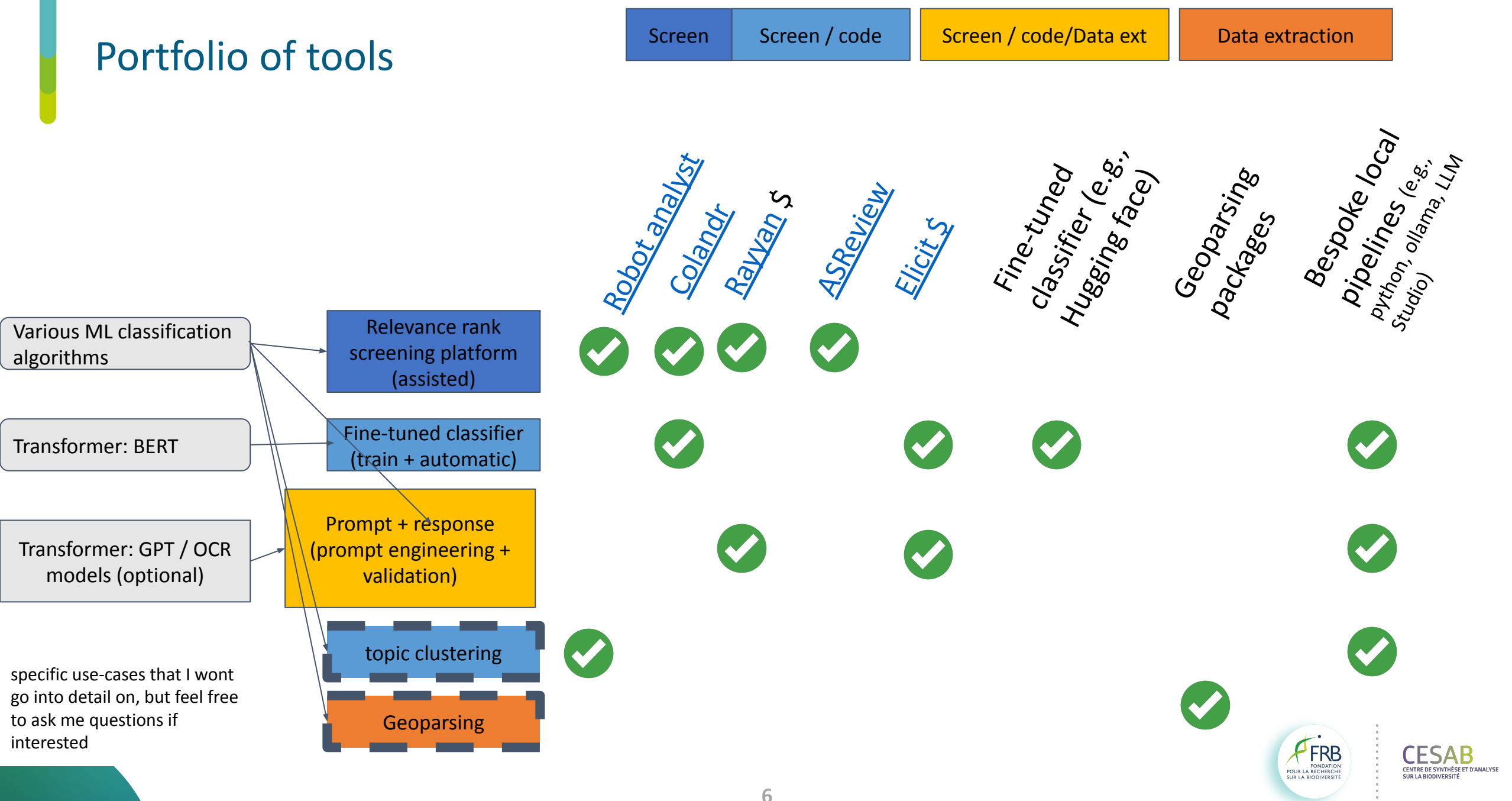
Need for machine learning literacy in evidence syntheses community

No substitute for domain-specific expertise, but literacy in ML methods is also needed.

- While there are increasing numbers of online platforms that make ML models user-friendly without need to code, still limited in how customizable they are.
- Without experience, researchers may have unrealistic expectations of what the ML system can achieve
- Needed for assessing whether a model/tool is fit for purpose.



Portfolio of tools





Portfolio of tools

For each of these categories

Screen	Screen / code	Screen / code/Data ext
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Break them down into the following steps:

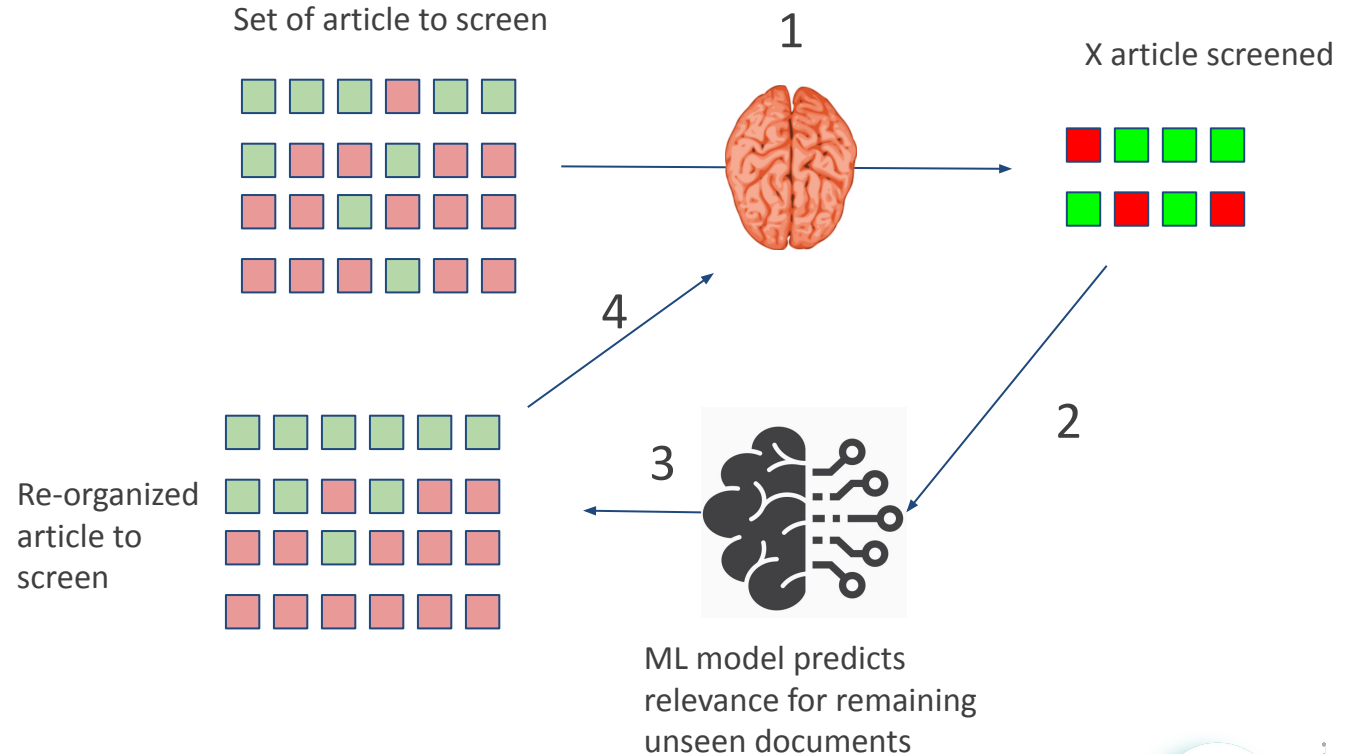
1. Concept (under the hood of how it works)
2. Application
3. Tools
4. Validation
5. Summary of strengths and limitations

Relevance rank screening platform (assisted)

Screen

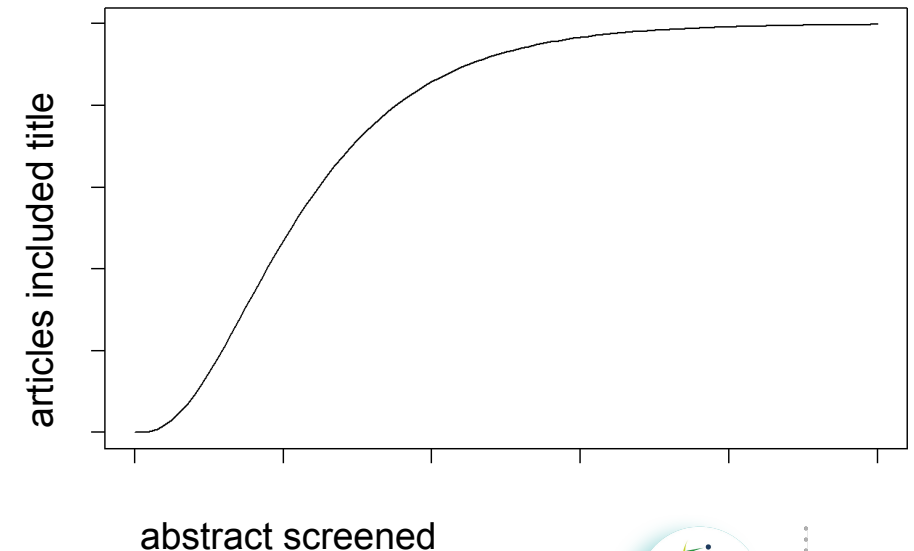
Concept: After you screen/code a certain number of articles, the underlying ML algorithm learns from that training data to predict the relevance of 'unseen' documents.

How they work: various ML algorithms used depending on the platform



Application:

- Truncate screening. As you screen, the model continually trains itself to recognize relevant articles. It will then continually re-order the remaining unseen documents so the most likely to be included will be shown first. (e.g. [Cohen et al 2009](#) ; [Apriyani et al., 2024](#)). Time advantage = you can stop screening early once you hit a pre-defined stopping criterion.
 - threshold e.g. 5% in [Cheng et al., 2023](#)
 - Asymptote e.g. [Rubenstein et al., 2023](#)
- Can be used to update reviews: e.g. [Cohen et al 2005](#) , [Cohen et al 2009](#)



Relevance rank screening platform (assisted)

Screen

Tools:

- many screening platforms, e.g.: ASReview, Colandr

Validation

- False positives are controlled in the pipeline as humans are in the loop. Minimizing the false negative rate is also built in to the criteria defining the stopping point

Strengths	Limitations
• Risk of bias is low	• Humans still need to screen all included articles (depends on corpus size) • Can't know definitively a priori how many articles will be included and thus how many you will need to end up screening. (but you can estimate in a scoping exercise)

Detour: Modern large language models explainer



For understanding the next set of tools:

Screen / code

Screen / code/Data ext

we need to take a quick detour

Detour: Modern large language models explainer



What is a large language model?

- probabilistic (predictive) models. They do not ‘understand’ text, but rather are very advanced statistical machines ([c.f. paper on the transformer model architecture](#)) that make predictions after being trained on large amounts of text.



Detour: Modern large language models explainer

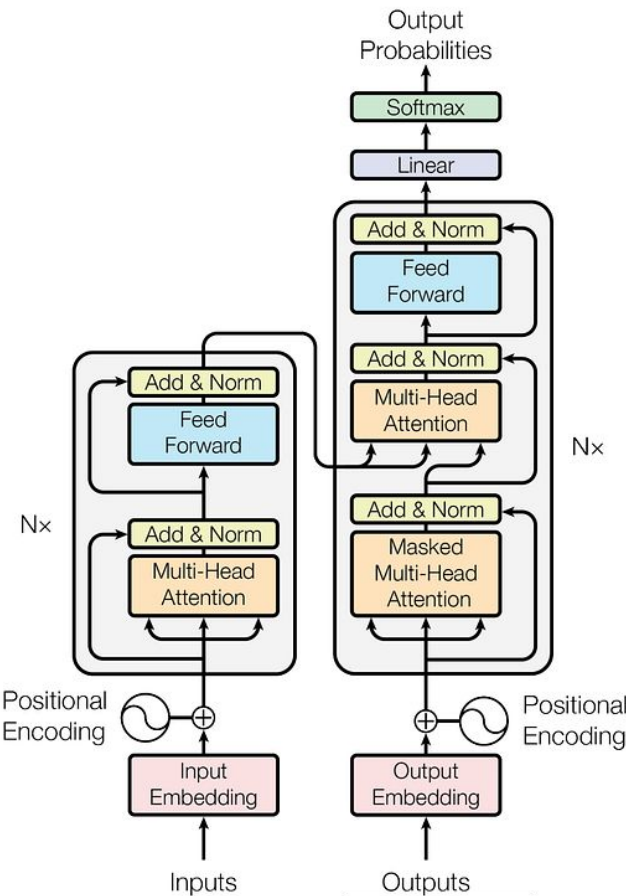


How do they work?

Text classification, where the model must classify a piece of text into one of several predefined categories.

The encoder takes in a sequence of tokens and produces a fixed-size vector representation of the entire sequence, which can then be used for classification.

BERT
Encoder



Language generation, where the model must generate a sequence of words based on an input prompt or context.

The decoder takes in a fixed-size vector representation of the context and uses it to generate a sequence of words one at a time

GPT
Decoder

Detour: Modern large language models explainer

Step 1: Represent text numerically

1.1: Tokenisation

Take raw text, convert it into tokens (numbers). [See ex](#)

The CESAB literature review course is awesome

Clear Show example

Tokens	Characters
8	45

The CESAB literature review course is awesome

[976, 105985, 2912, 23216, 3358, 4165, 382, 15339]



Sequence of words



Integers



Detour: Modern large language models explainer

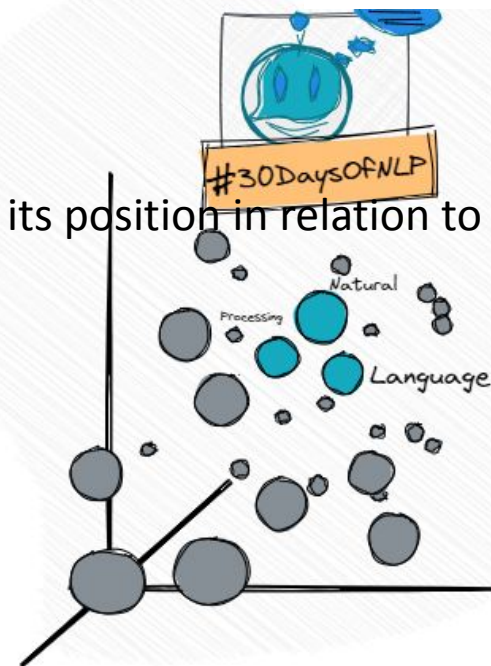


Step 1: Represent text numerically

1.2: Input embedding + positional encoding

For each token, you give the model a vector of numbers to train, as well as information about its position in relation to the other words in the sentence.

- Each token (integer) is accompanied by a vector
- A sequence of vectors → an embedding matrix where the i -th row of the embedding matrix = the embedding of the i -th token (this is a trainable parameter).
- At the end of each vector, adds information on position of the token in the sentence



Why: to train matrix that represents high-dimensional vector space where semantic relationships between tokens become mathematical relationships

Detour: Modern large language models explainer

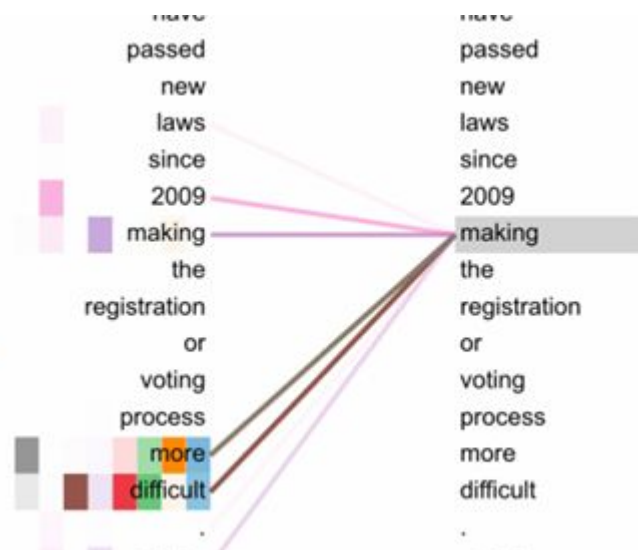


Step 2: Attention

The model uses these inputs to calculate the relative meaning of the word inside the sentence, because depending on the context, the same word can have different meanings

*‘more’ and ‘difficult’ contribute to the word ‘making’.
So rather than its generic meaning, it’s related to
making something more difficult.*

Take home message: These models do not ‘understand’ text. Nor can you think of understanding their behaviour in terms of keywords, as words can take on varying meanings depending on context. Best to think of them as a statistical black box where the mechanisms are mysterious. Therefore it is important to **VALIDATE/AUDIT** their output



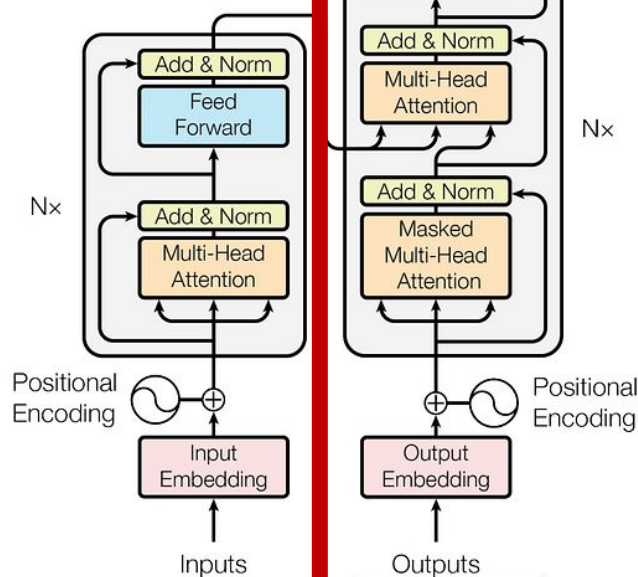
Encoder block for classification

Screen/code

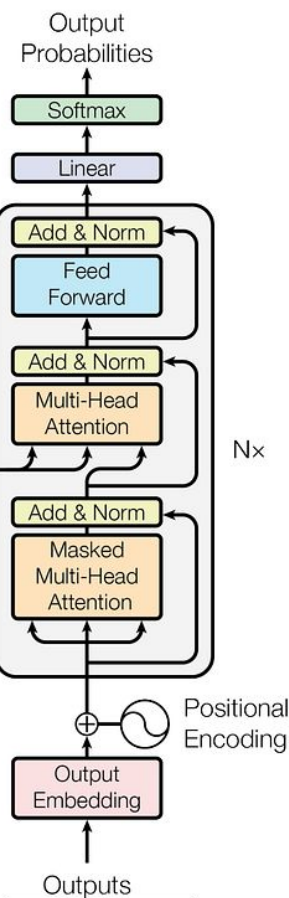
Text classification, where the model must classify a piece of text into one of several predefined categories.

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BERT
Encoder



GPT
Decoder



Encoder block for classification

Screen/code

Applications:

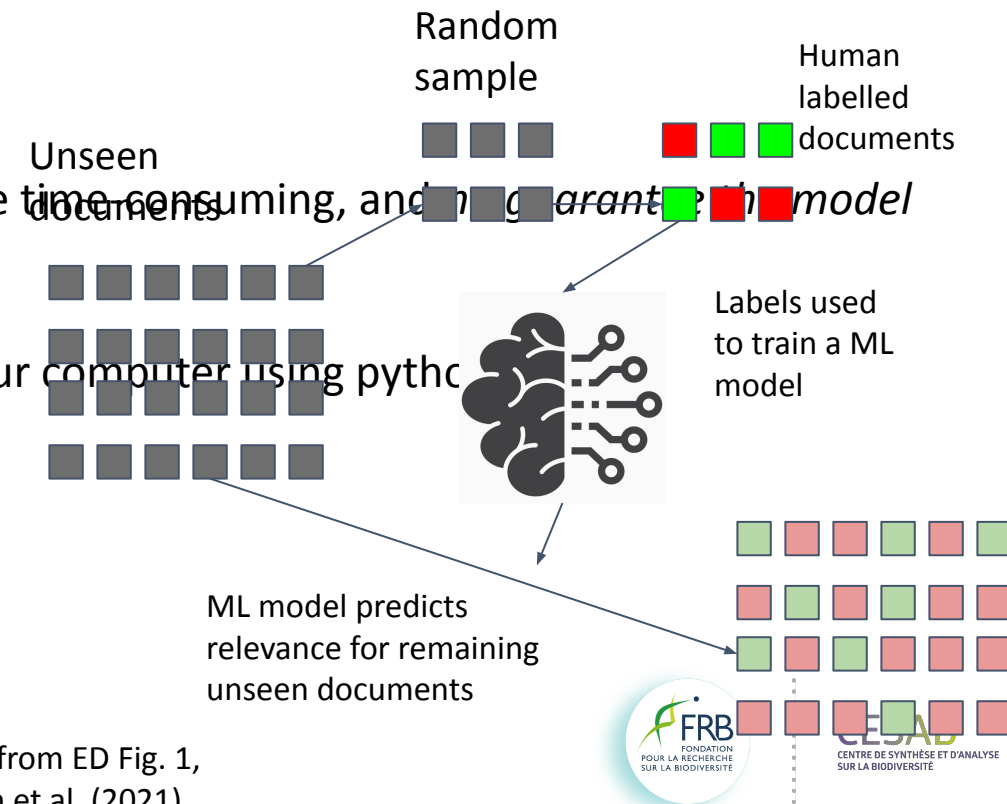
Screening → training the model to classify unseen text as include vs exclude. [similar to relevance ranking, except this learns from training data, and then predicts classifications automatically]

Categorical coding (classification) → training the model to classify included articles across descriptive metadata categories/typologies (e.g., study type, intervention type, etc).

Requirements:

Need to provide it with a human-labelled dataset to learn from (can still be time-consuming, and no guarantee *model will perform well*)

Tools: Some platforms may use these models, or can train your own on your computer using python



Encoder block for classification

Screen/code

Validation

Step 1: Divide data for cross validation

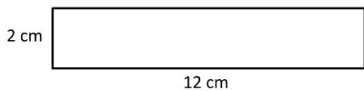
E.g. if you had 100 math questions and their answers, and you go through the questions and answers for some of them, and then you test your performance after

Name

Date

MENTAL MATHS SHEET 4:9 ANSWERS



1)	Write down the number four thousand and sixteen.	4016
2)	$154 + 30$	184
3)	How many tens in 180?	18
4)	Round 828 to the nearest 100.	800
5)	Half of 72	36
6)	60 subtract 17	43
7)	What is the perimeter of this rectangle? 	28cm
8)	$4000 + 300 + 5 =$	4305
9)	$24 \div 8$	3
10)	What digital time is shown on the clock? 	
11)	What is the remainder when 22 is divide by 5?	
12)	A train journey takes 1 hour 40 minutes. If I set off at 9:30am what time would I arrive?	
13)	How many meters is $3 \frac{1}{2}$ kilometres?	
14)	The temperature on a cold day is -3°C . It rises by 5°C . What is the temperature now?	
15)	$\frac{3}{8} + \underline{\hspace{1cm}} = 1$	
16)	I am a square with sides length 4cm. What is my area?	

probably work
yourself and check



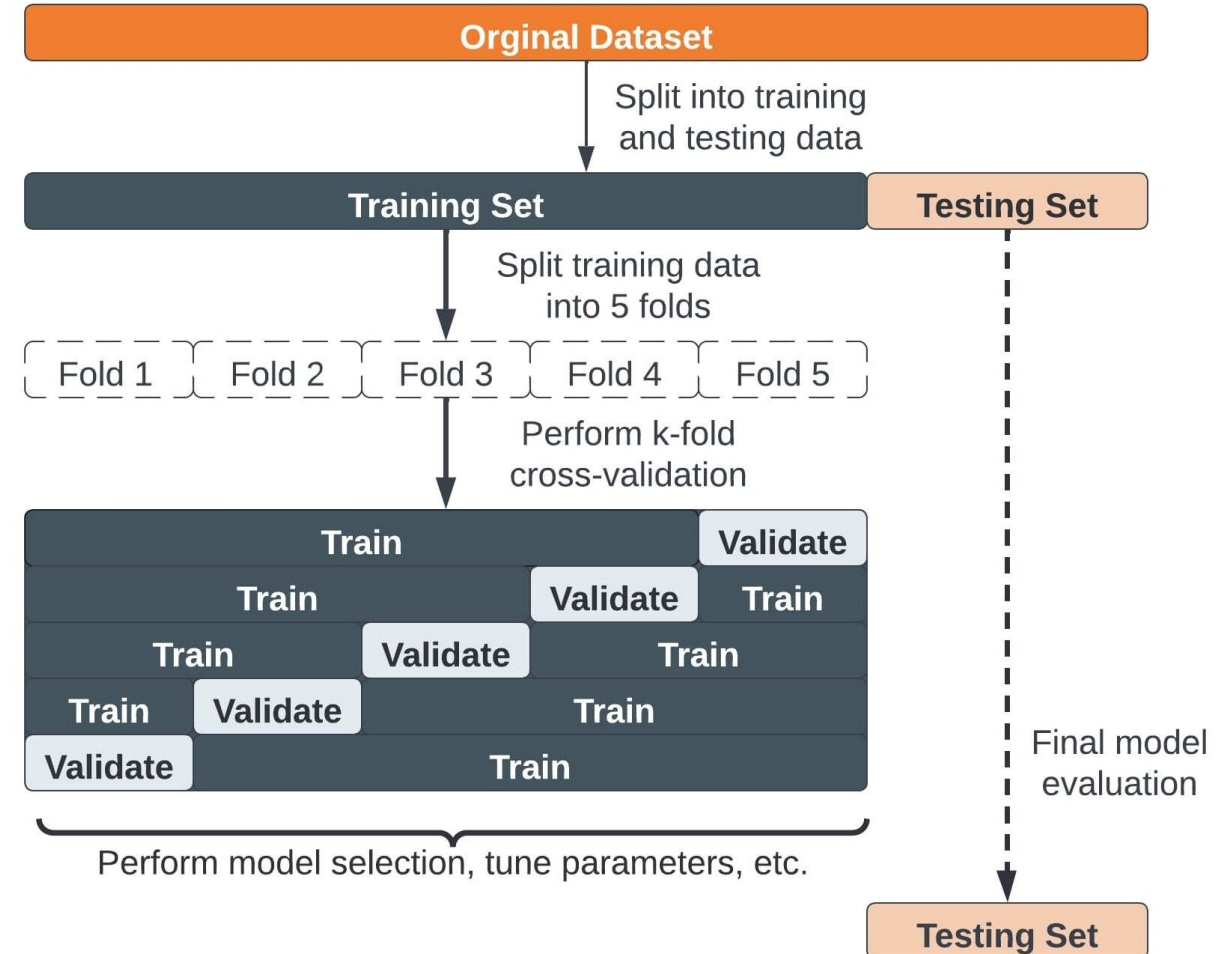
Encoder block for classification

Screen/code

Step 1: Divide data for cross validation

- We divide this data into a training set and test set.
- Use the test set to train the model
- Use the model to make predictions for the test set, and

ideally you want enough data so you can perform k-fold cross of the training/test set).

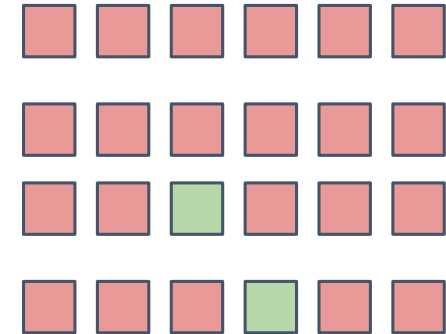


How to we evaluate the model?

Accuracy: Out of all the predictions we made, how many were true?

Example: We have a data set with 100 articles, and 2 are relevant. Our model could predict every document as irrelevant and it would have 98% accuracy, but it would be a terrible model for our purpose (finding the relevant articles)

So we need a way to maximize our *recall* of relevant papers, while avoiding including too many irrelevant papers (*precision*)



Encoder block for classification

Screen/code

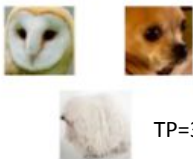
How to we evaluate the model?

Recall: how good the model is at finding all the relevant articles (TP/TP+FN)

Precision: Out of all the articles included, how many are truly relevant (TP/TP+FP)?

Generally aim for $F1 > 0.7$ for simple (binary) classification problems. For more complex multi-label, F1 scores can be as low as 0.45. But, this implies a larger amount of error – deciding whether a model is fit for purpose depends on the purpose. For a map this is probably sufficient to describe general patterns in the data, but probably not for a rigorous meta-analysis.

F1 Score: combines recall and precision. Because there is a trade-off between precision and recall, F1 measures how the model makes that trade-off.

		Predicted	
		Animal	Not animal
Actual	Animal	 TP=3 FN=0	
	Not animal	 FP=1	 TN=2

Model 4

Accuracy	83%	$\frac{3+2}{3+2+0+1}$
Precision	75%	$\frac{3}{3+1}$
Recall	100%	$\frac{3}{3+0}$
F1 score	86%	$2 \cdot \frac{0.75 \cdot 1}{0.75 + 1}$

Screening & coding classifiers

ARTICLES

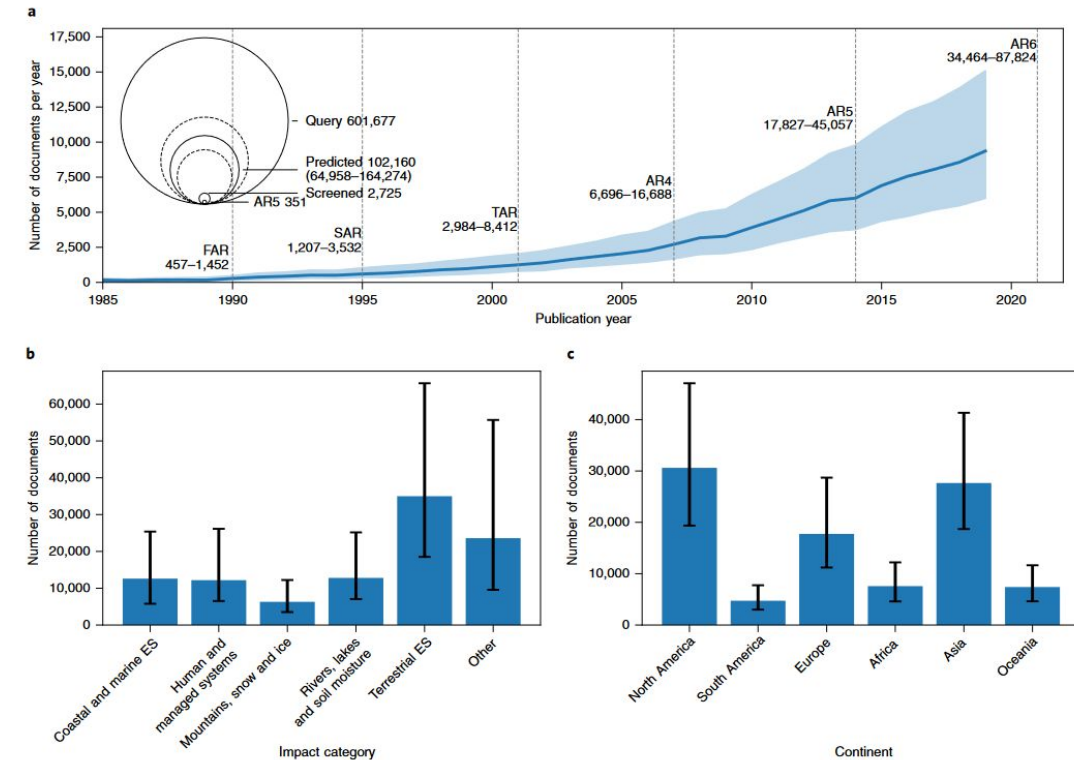
<https://doi.org/10.1038/s41558-021-01168-6>

nature
climate change



Machine-learning-based evidence and attribution mapping of 100,000 climate impact studies

Max Callaghan ^{1,2} ✉, Carl-Friedrich Schleussner ^{3,4,5}, Shruti Nath ^{3,6}, Quentin Lejeune ³, Thomas R. Knutson ⁷, Markus Reichstein ^{8,9}, Gerrit Hansen ¹⁰, Emily Theokritoff ^{3,4,5}, Marina Andrijevic ^{3,4,5}, Robert J. Brecha ^{3,11}, Michael Hegarty ³, Chelsea Jones ³, Kaylin Lee ³, Agathe Lucas, Nicole van Maanen ^{3,4,5}, Inga Menke ³, Peter Pfleiderer ^{3,4,5}, Burcu Yesil ³ and Jan C. Minx ^{1,2}



Encoder block for classification

Screen/code

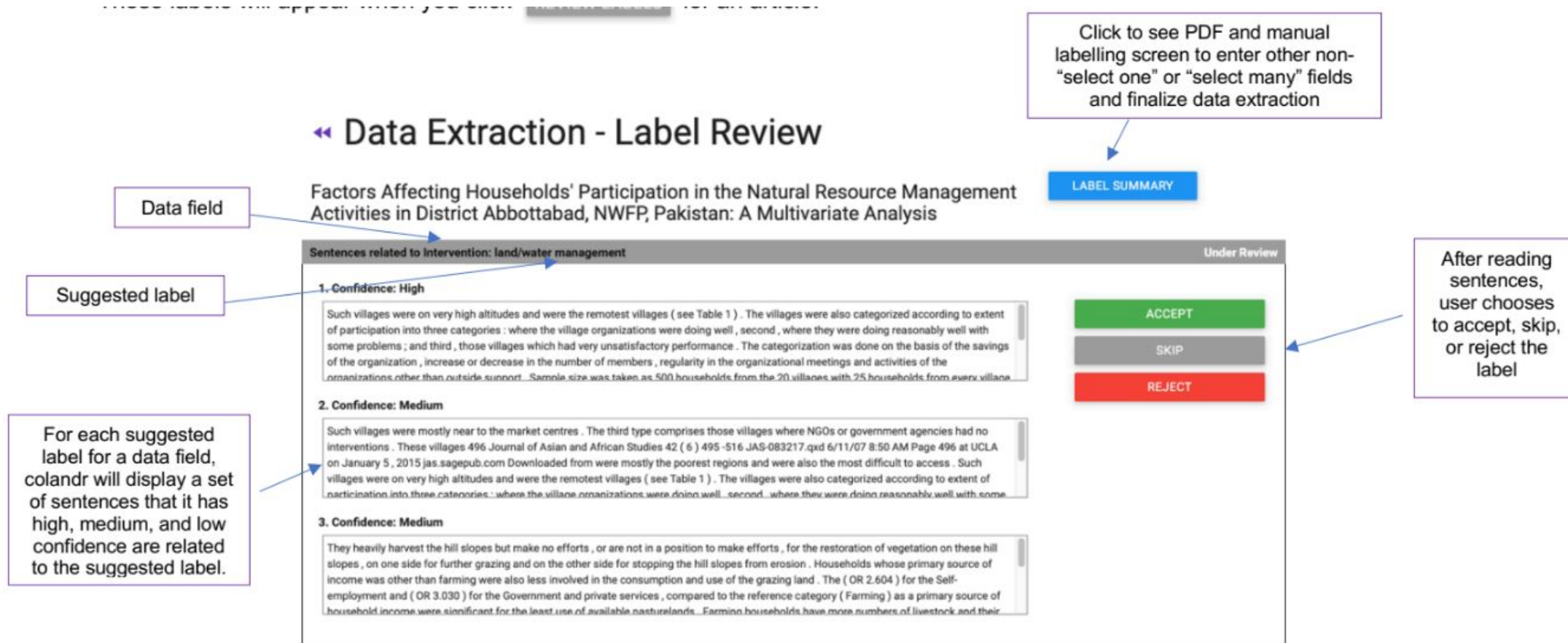
Alternative: Using pre-trained model

You can train your own model, or use pre-trained models off the shelf – applying existing models for predicting classifications on [Huggingface](#) to predict labels for unseen data (e.g. sentiment analysis, location extraction, etc.)

Encoder block for classification

Screen/code

Provide suggestions to expedite data extraction (e.g. [Colandr](#))



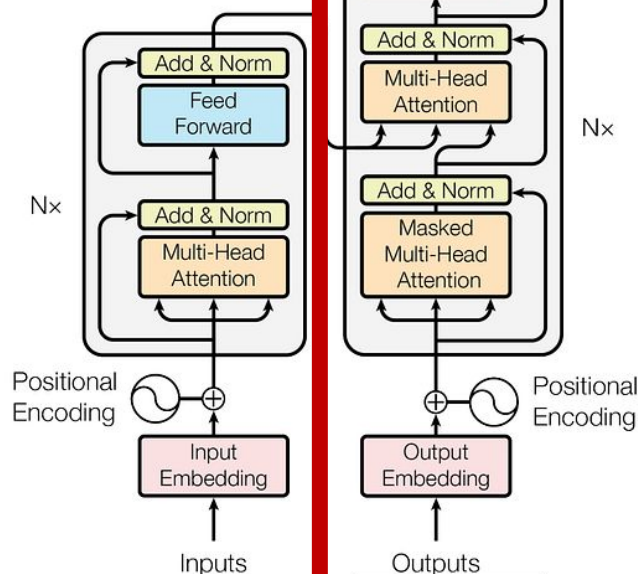
Strengths	Limitations
• Can describe large corpuses of evidence that would otherwise not be possible	• Human labelling + training the model still takes time & expertise • No guarantee that the model will perform well • Risk that some areas of the literature are not sufficiently present in the training sample, which would lead them to be under-represented in the final outcome. • Hidden biases... • Considerations when using for metadata classification ○ Generally struggle with nuanced delineations, or typologies with many categories. To be safe, try organising typologies hierarchally, so labels can be re-grouped and simplified if needed ○ Each label needs sufficient data to fit the model. So if some labels have poor representation, they are less likely perform well.

Decoder block: Prompt + response

Screen/code/Data ext

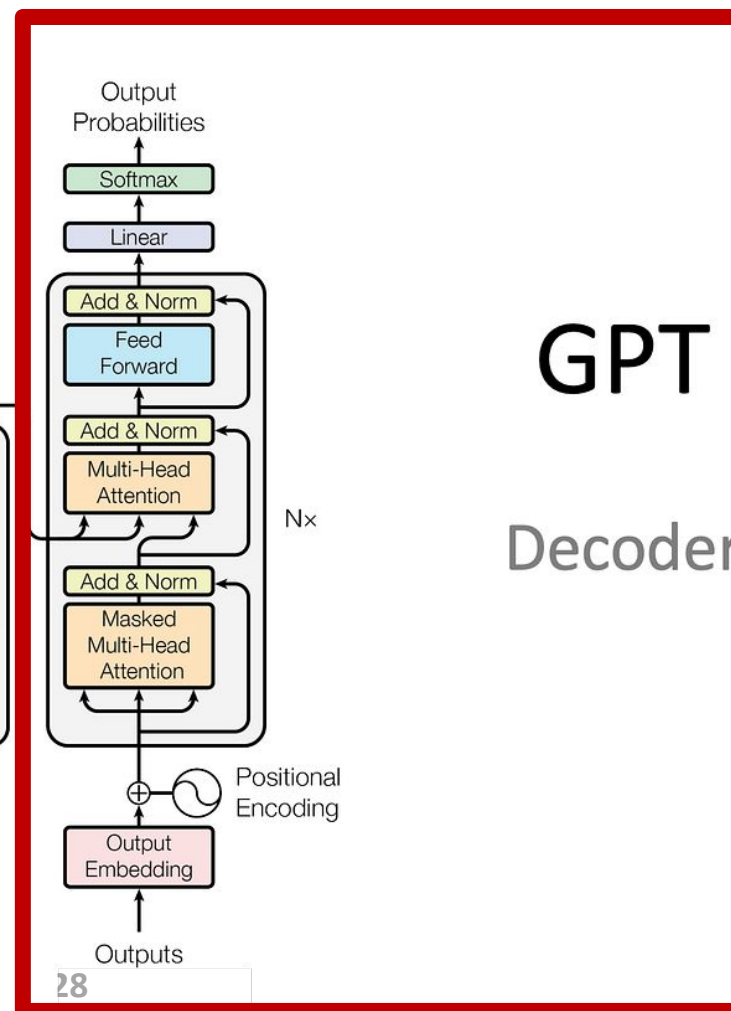
BERT

Encoder



GPT

Decoder



Language generation, where the model must generate a sequence of words based on an input prompt or context.

The decoder takes in a fixed-size vector representation of the context and uses it to generate a sequence of words one at a time

Decoder block: Prompt + response

Screen/code/Data ext

A decoder is almost the same structure as the encoder → designed to predict the next word in a sequence given the previous words. [See this illustration](#)

To predict the next word (token):

- Each possible token is assigned a probability according to a probability distribution.
- The LLM temperature parameter modifies this distribution.
 - A lower temperature makes the tokens with the highest probability more likely to be selected (ie. coherence)
 - a higher temperature increases a model's likelihood of selecting less probable tokens (more creative)
 - Different temperature settings introduce different levels of randomness and allow you to balance how the model performs.

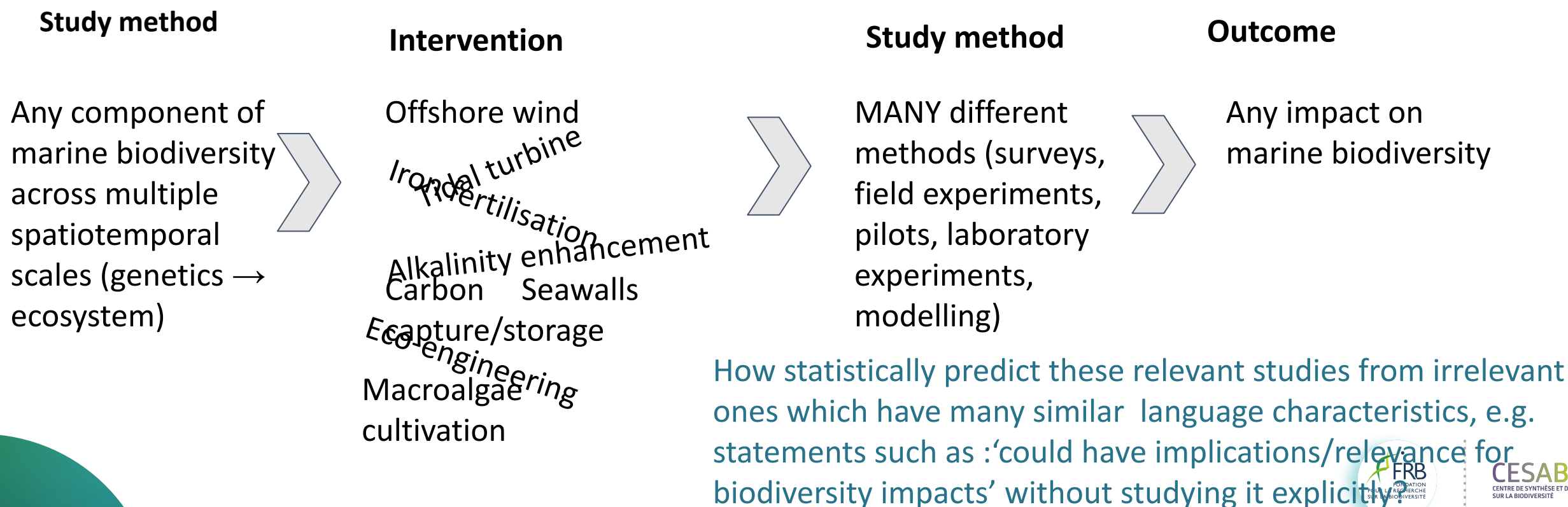
Therefore, it is important to recognize that each time you run the model, your results might change due to randomness → c.f. validation

Decoder block: Prompt + response

Screen/code/Data ext

Applications:

- Complex coding/classifying tasks where a classification model performs poorly: Screening for articles on biodiversity impacts from ocean-based solutions

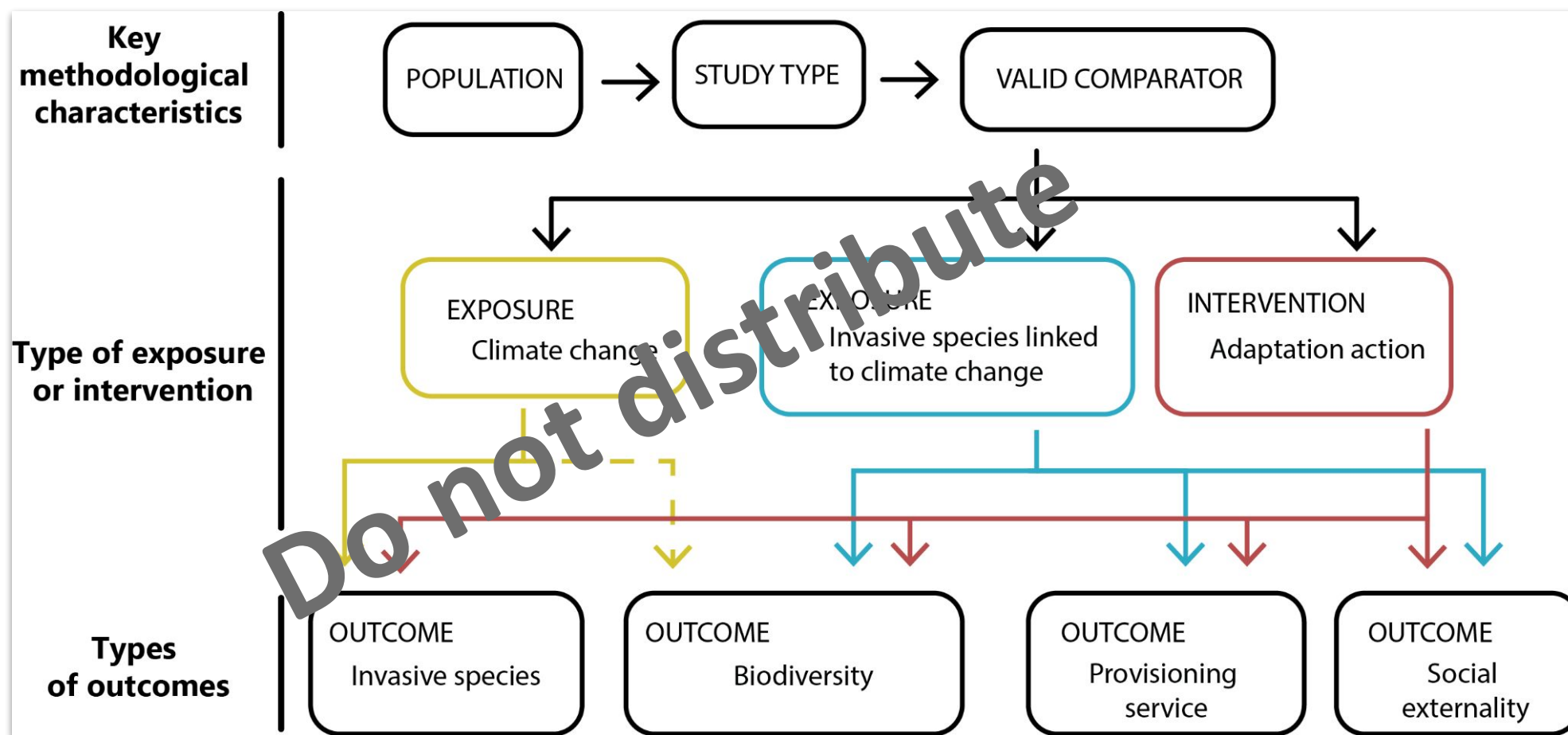


Decoder block: Prompt + response

Screen/code/Data ext

Applications:

- A logic/rule based Q&A approach makes sense for your research question. Eg. research on invasives linked to climate change



Decoder block: Prompt + response

Screen/code/Data ext

Applications:

- Extracting data:

Prompt: “what is the name of the study site, and the year the study was conducted?”



Output: formatted response

```
{  
  site: [Hawaii],  
  year: [2015]  
}
```

Source text: *The study was conducted in Hawaii in 2015.*

shutterstock.com • 2655333777

Decoder block: Prompt + response

Screen/code/Data ext

Requirements: No training data needed, but you need to spend time evaluating your model responses against human labels/extractions, and quantifying model performance.

Tools: Different tools available with different strengths/weaknesses:

	User friendly	Model size	Open-weight	Data security	Cost effective (consider iterative prompt development)	User control (integration validation pipelines)	Time
paid online platforms (e.g., Elicit).	✓	✓					✓
bespoke code* with paid model API		✓				✓	✓
bespoke code* with free open-weight models			✓	✓	✓	✓	

* Tools for bespoke code: python, LLM studio, etc.

Decoder block: Prompt + response

Screen/code/Data ext

Validation: There is no one structured method to quantify performance, as it will depend on the task (ie. screening/binary choice; classification across multiple labels; quantitative data extraction).

There are different possible types of error to consider with different severities.

For example in a multiple label task, an LLM may provide an additional label compared to the human reviewer, yet when assessed by a human reviewer the additional label is sensible. Conversely, the LLM may miss a label identified by a human but get another right, or get one label right and add an incorrect label. Furthermore, how does the LLM performance compare with the inter-reviewer variation across these different error types?

In a quantitative example, such as geoparsing, error quantification may want to consider the Haversine distance between the geoparsed coordinates and those in text.

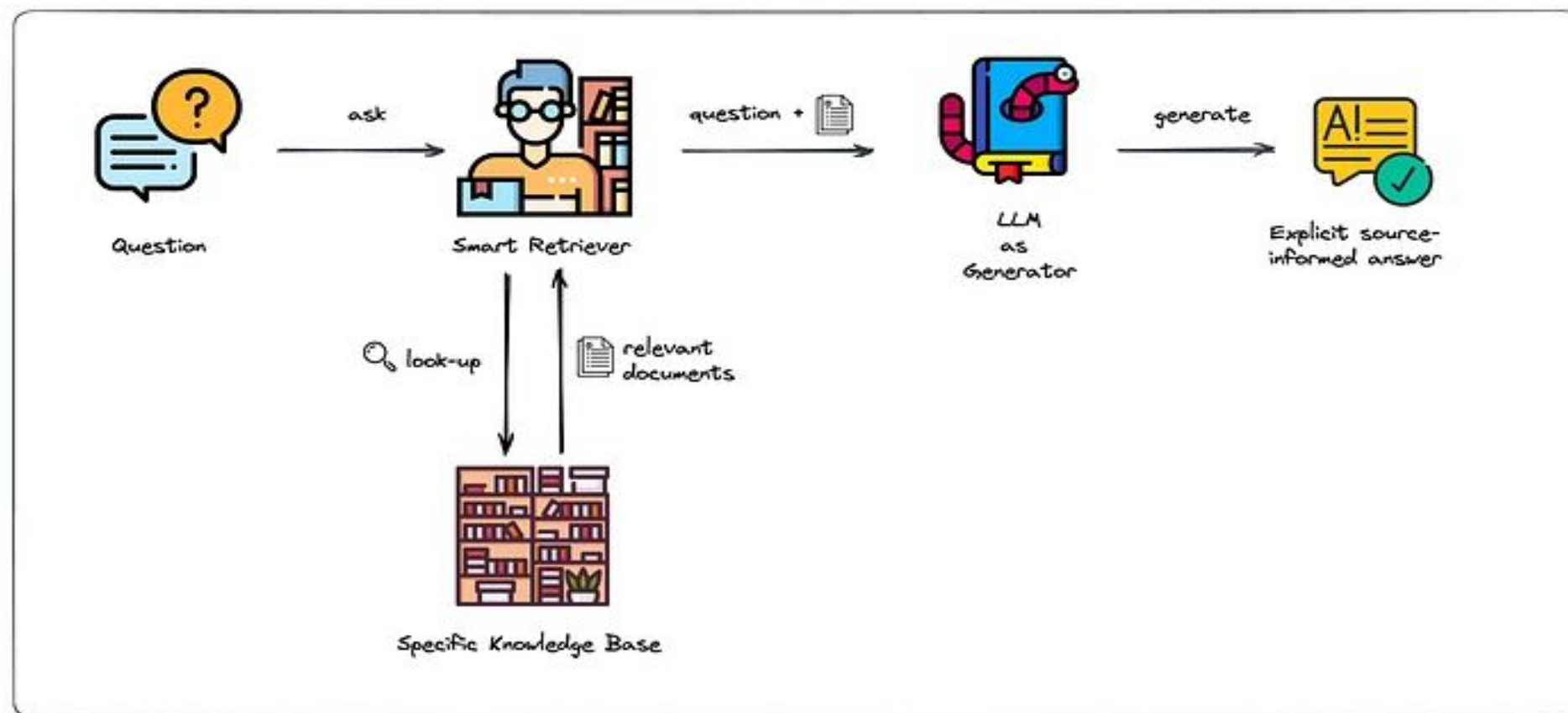
Stochastic. you may get different answers even running the same prompt with the same context. You may need to check for response consistency

Check for hallucination

Decoder block: Prompt + response

Screen/code/Data ext

Example of implementation: Retrieval-Augmented Generation (RAG) allows the user to pose a query (prompt) to an LLM, which can then reference a specific knowledge base (context) outside of its training data sources before generating a response. This allows you to ask an LLM to answer questions about a source text.



Decoder block: Prompt + response

Screen/code/Data ext

Now let's see an example demonstration of what this looks like in code...

Decoder block: Prompt + response

Screen/code/Data ext

Strengths	Limitations
• Can *potentially* answer more nuanced questions, extract certain types of data.	• Does not necessarily save time, as iteratively designing prompts and validating at each step for adequate performance is VERY time-consuming • Different models and tools have different limitations – it is important to choose the right approach for your question and validate ^ • Model responses can reflect biases in the training data – importance of validation ^^ • Current LLMs are trained on a wide range of text data, and do not have domain-specific knowledge. So treat it like assigning a 10-year old to do your data extraction. Answers that may seem obvious to you, might confuse the LLM (doi: 10.1038/s44185-024-00043-9; 10.1016/j.ecoinf.2026.103699)

Ethics: Biases

LLMs are statistical ‘black boxes’ and therefore it is difficult to know *where* they get their predictions from. This means that they may propagate biases of the text data they were trained on in unexpected ways, and it is important to *audit* these models to see if biases emerge that could affect the validity of using the model for your research question.

Examples of potential bias using the ClimateBERT language model to fill in the blank, from: [10.1038/s41558-023-01890-3](https://doi.org/10.1038/s41558-023-01890-3)

Prompt	Most likely words	Bias
Climate change adaptation [blank] women	for (34.5%) by (13.6%)	Women are seen as victims who are recipients of adaptation efforts rather than actors with agency ^{91,92}
Climate change adaptation [blank] men	by (27.7%) for (23.3%)	
Adaptation in the USA is [blank]	underway (15.0%) ongoing (9.0%)	The focus in Bangladesh is on the vulnerability and the need for more action, while the USA is depicted as a place where adaptation is already happening
Adaptation in Bangladesh is [blank]	critical (10.9%) urgent (10.5%)	

Ali [blank] climate change	denies (9.1%) blamed (6.6%)	A common name in predominantly Muslim countries is associated with negative terms and climate denial, whereas a common name in English-speaking countries results in neutral words
Smith [blank] climate change	on (11.1%) discussed (7.2%)	
The task was given to the project leader; [blank] completed it	he (49.1%) they (21.0%)	Project leaders are assumed to be men more often than other genders ('she' scored a probability of 2.5% in the first example, while 'they' scored 3.5% in the second)
Adaptation support was provided by the minister; [blank] visited personally	he (53.1%) she (10.3%)	



Ethics: Data ownership/privacy

Copyright and IP:

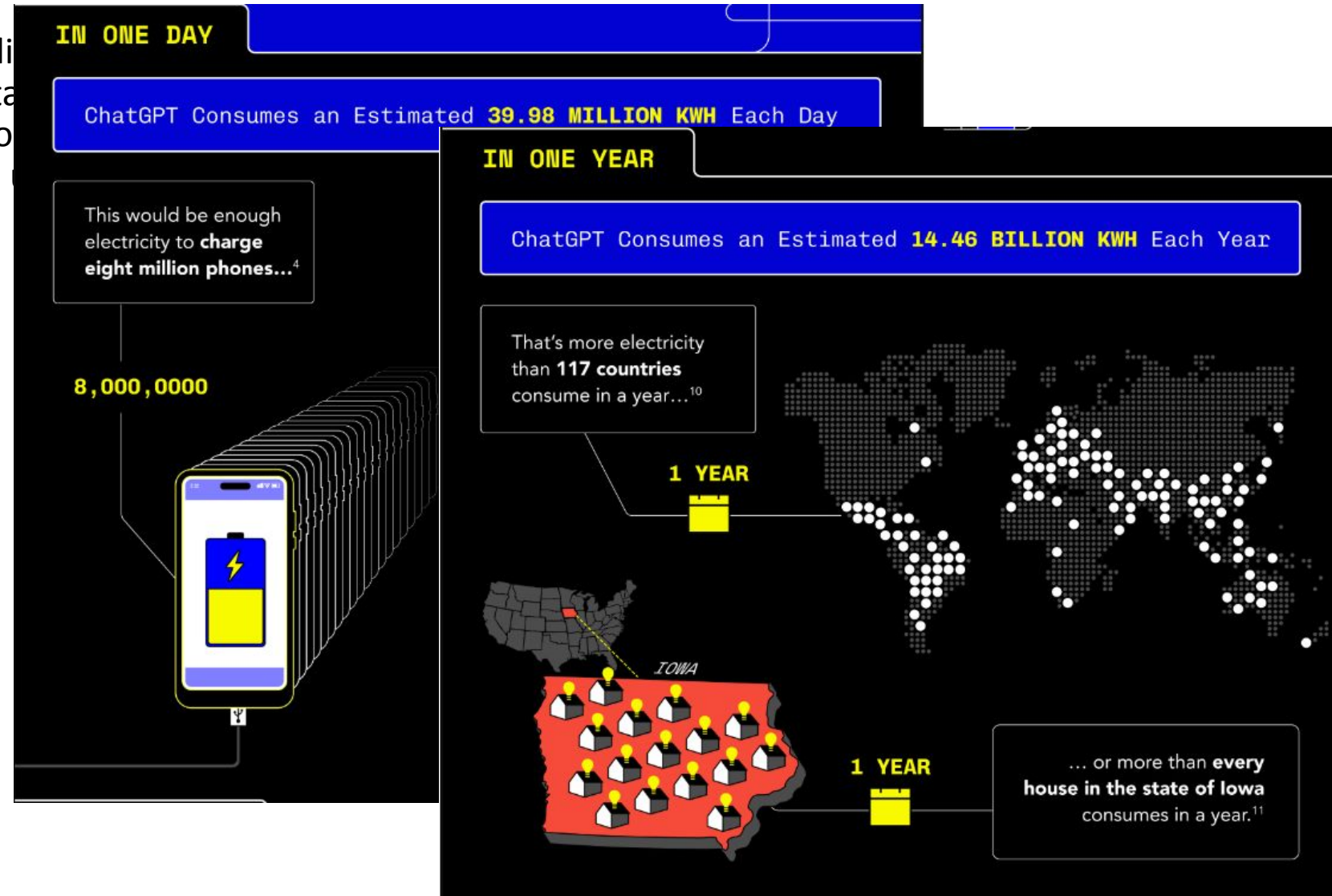
- ensuring you do not breach copyright, license agreements or infringe thirdparty intellectual property reuse when uploading content to AI systems
- protecting the IP of your own work when using AI systems (not only is this important for your own work, but if you unintentionally assign rights to your work to the AI system, it could compromise your ability to publish the work at a later date).
- What consider: Check the individual regulations around the tool you're using, and the safest is always to use a local model (e.g., DeepSeek on your computer)

Privacy

Be cautious about whether you can upload sensitive content, such as private or personal information, content under confidentiality agreements or embargoes

Ethics: Climate impacts

- Extraction of raw materials (including rare earth minerals)
- Supporting infrastructure (e.g. data centers)
- The energy-intensive phases of model training and inference
- Energy + water when the model is used





Take-home message

Not: Don't use AI to do evidence synthesis.

Instead: AI can be used, provided you can demonstrate that its use does not compromise the methodological rigour or integrity of the synthesis.

Make sure your method is fit for purpose.

- *Within the context of your research question (purpose):* justify its use, and validate its performance. What decision is the AI making, and how consequential is an error? Does the AI perform well enough for this particular task, research question and evidence base?
- Declare your LLM use in your method transparently.
- When interpreting results, consider sources of error and bias. I.e. don't over-interpret/overstate.



Exercise: Application of RAISE guidance

AI use is acceptable with human oversight, but any AI that makes or suggests judgements needs to be transparently reported.

- Justify why an LLM was used to extract data, and briefly state/describe applicable biases and limitations.
- Which LLM was used, its version, and what were its settings? What information is available (closed or open weights, what were the training data)
- Why is the AI tool selected suitable for your specific task and context? Was it previously used effectively in similar methods? If so, cite these works.
- Describe whether when iteratively designing and testing prompts, was this done within the same ‘conversation’ with the LLM, or was a new conversation used for each new attempt? All of this can affect performance and replicability.
- What was the reference/gold standard? Was human disagreement considered? Sample size? Does the training/validation data of reflect the variability of the domain of interest?
- Are the final prompts used provided as a supplementary file?
- How is model performance evaluated? (e.g., metrics, FP, FN, different types of error, stability/performance across runs)
- Was human oversight used? If so how, and if not, is it justified?

c.f. **Responsible use of AI in evidence SynthEsis (RAISE)**: recommendations and guidance (doi: 10.17605/OSF.IO/FWAUD)



Exercise: Application of RAISE guidance

In sub-groups, you will be given an article which uses an AI tool in an evidence synthesis.

[Groups and paper assignments in your links for participants document](#)

Act as the reviewer. Evaluate the article's use of AI and write a Reviewer report.

Consider:

- What did the Authors do well?
- What is missing?
- Do their methods make sense for their research question?
- Do their methods support their conclusions? Are claims supported?

There is no such thing as a 'perfect' study. So the point is not to say research is bad, but rather to evaluate the method in context and determine whether the findings support the interpretation/claims, and if not, why.